An End-to-End Framework for Urban Planning Analysis from Street-View Imagery

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Table of Contents

1. Introduction
   - Background and Motivation
   - Problem Definition
2. Proposed Framework
   - Overview
   - Stages 1–2: Privacy and Georeferencing
   - Stages 3–4: Vision
   - Stages 5–6: Metrics and Mapping
3. Results and Discussion

Table of Contents

1. Introduction
   - Background and Motivation
   - Problem Definition
2. Proposed Framework
   - Overview
   - Stages 1–2: Privacy and Georeferencing
   - Stages 3–4: Vision
   - Stages 5–6: Metrics and Mapping
3. Results and Discussion

Street-View Analysis Depends on Commercial Imagery

- Street-view imagery (SVI) is a primary source for urban analysis: streets, buildings, vegetation, infrastructure.
- Most pipelines rely on Google Street View or Mapillary - georeferenced, but with uneven coverage in under-represented regions.
- Dashcam video is cheap and flexible, but arrives raw: identifiable people, GPS burned into the frame, redundant and blurred frames.



Jeddah dashcam frame and its segmentation

Consequence

Existing frameworks treat preprocessing, segmentation and metric extraction as isolated steps and assume pre-georeferenced input, so raw video cannot be used directly.

Research Gap and Contributions

Three practical barriers

(i) Privacy
faces and plates; detectors trained on Western plates

(ii) Georeferencing
GPS exists only as a text overlay in the video

(iii) Fragmentation
indicators used individually, no unified score

1. End-to-end pipeline

Raw dashcam video → georeferenced urban-quality maps.

2. Geographic domain adaptation

Region-specific licence-plate anonymisation.

3. Urban Visual Score (UVS)

Seven indicators, three newly introduced (TRF, TRN, HUM).

4. Cross-city validation

Jeddah, Frankfurt, Paris - no city-specific reconfiguration.

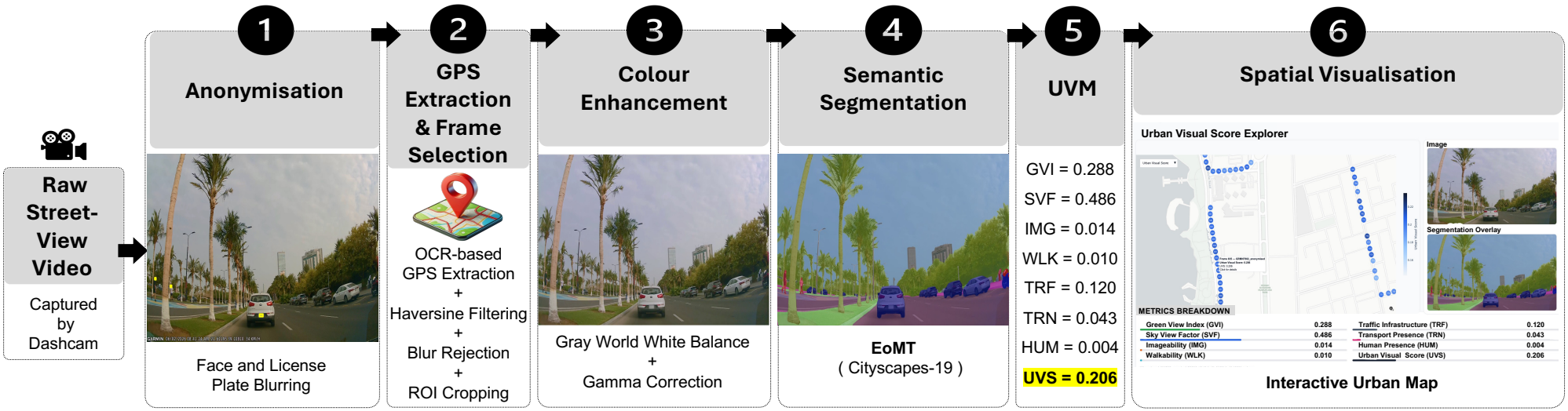
Table of Contents

1 Introduction
- Background and Motivation
- Problem Definition

2 Proposed Framework
- Overview
- Stages 1–2: Privacy and Georeferencing
- Stages 3–4: Vision
- Stages 5–6: Metrics and Mapping

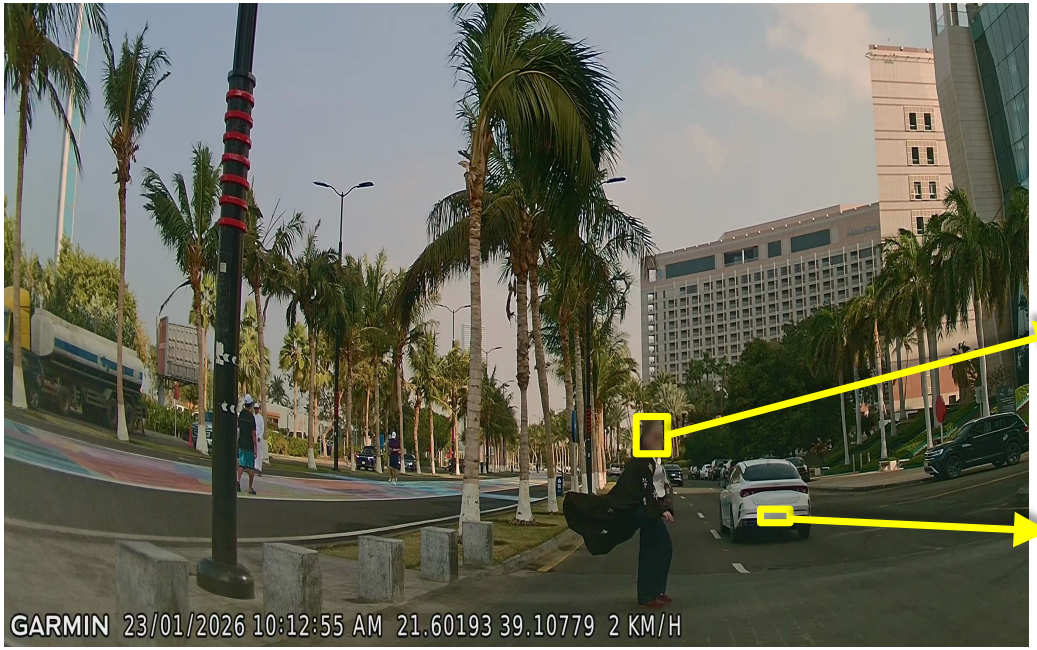
3 Results and Discussion

Six-Stage Framework



Raw dashcam video → privacy-preserving, georeferenced urban analysis.

Stage 1: Privacy-Preserving Anonymisation





Face Anonymisation

Face Detection → Gaussian Blur
YOLOv8n-face (tiled) + ByteTrack



Licence Plate (LP) Anonymisation

Vehicle Detection → Plate Detection → Gaussian Blur
YOLO12-based Pipeline + ByteTrack

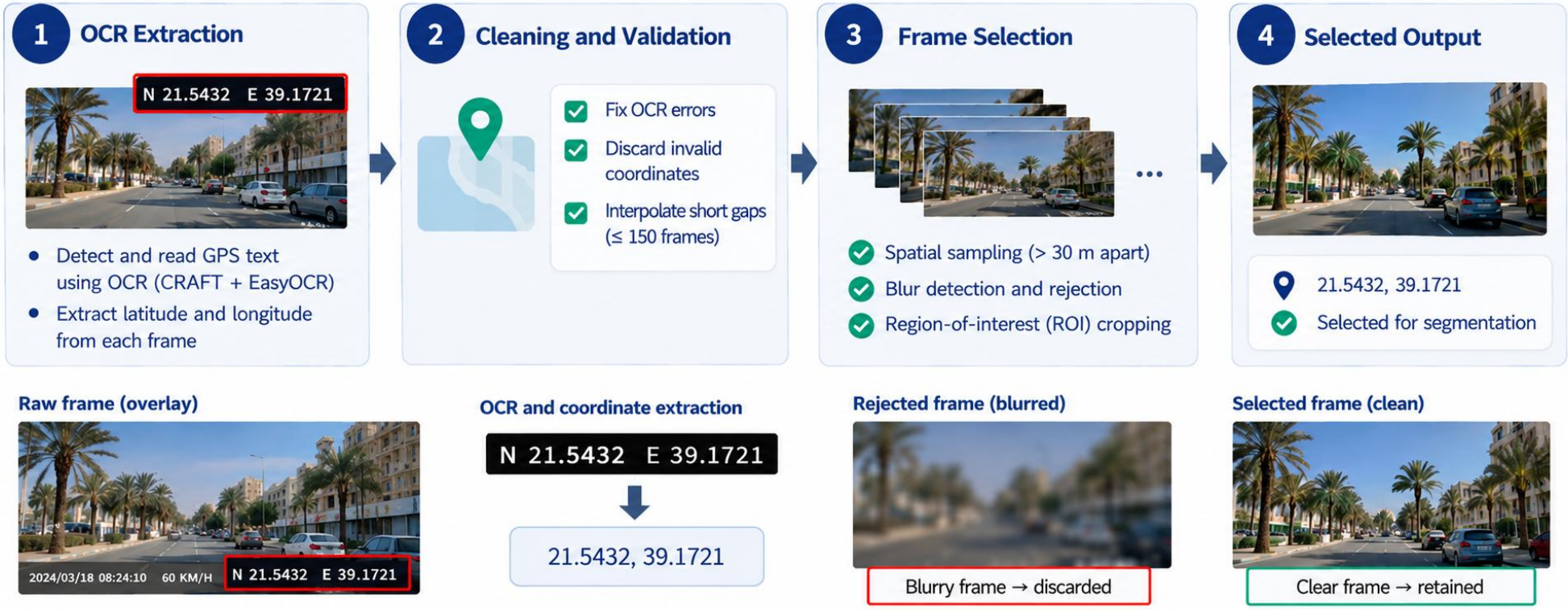
Faces and licence plates

Faces: YOLOv8n-face. Plates: vehicle detection followed by fine-tuned YOLOv12n. Gaussian blur applied.

Temporal consistency

ByteTrack with hold buffers (60 frames plates, 15 faces). Fallback: blur lower 12% of vehicle. No frames stored.

Stage 2: GPS Extraction and Frame Selection



Source: ChatGPT, AI-generated image (2026).

Stage 3: Colour Enhancement



Input Image

Illumination Inconsistency

Preprocessing



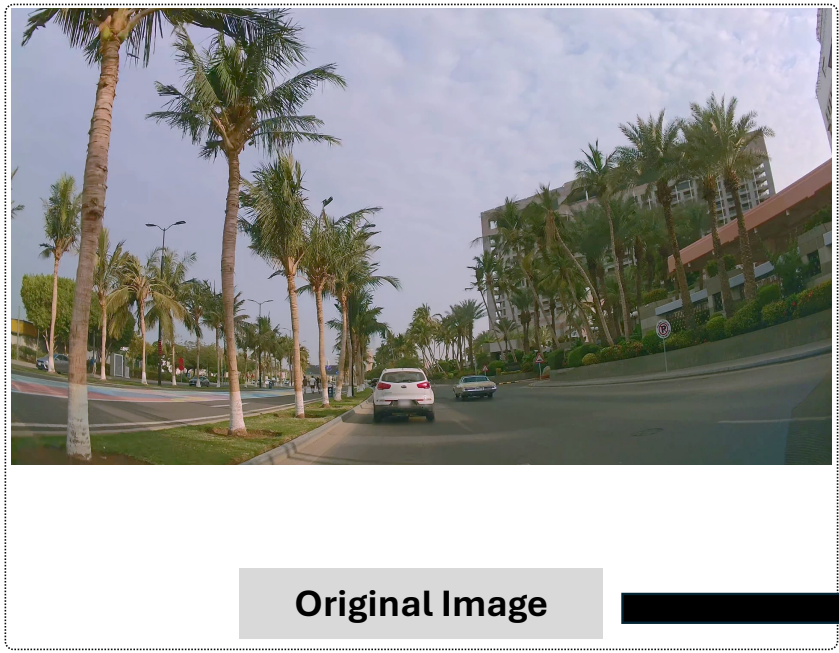
Colour-Enhanced Image

Gray World WB → Gamma Correction

Colour enhancement

Gray World white balance + gamma correction ($\gamma = 0.75$).

Stage 4: Semantic Segmentation



Semantic segmentation

EoMT with Cityscapes-19 weights,

used without fine-tuning.

Stage 5: Urban Visual Metrics and UVS

Pixel ratios over valid pixels: $X = \sum_{c \in X} N_c / N_{\text{total}}$

Indicator	Cityscapes classes	Weight
Green View Index (GVI)	vegetation, terrain	0.25
Sky View Factor (SVF)	sky	0.25
Walkability (WLK)	sidewalk, fence, pole	0.20
Imageability (IMG)	building, wall	0.15
Traffic Infrastructure (TRF)	sign, light, road	0.05
Transport Presence (TRN)	car, truck, bus, train, motorcycle, bicycle	0.05
Human Presence (HUM)	person, rider	0.05

Urban Visual Score (UVS)

$$\text{UVS} = 0.25\,\text{GVI} + 0.25\,\text{SVF} + 0.20\,\text{WLK} + 0.15\,\text{IMG} + 0.05\,(\text{TRF} + \text{TRN} + \text{HUM})$$

Orange = introduced in this work. Heuristic weights are fixed across cities; established indicators dominate, while context-dependent indicators contribute without dominating.

Stage 6: Spatial Visualisation



Urban Visual Score Explorer (Plotly Dash + OpenStreetMap)

Interactive Spatial Explorer

UVS-coded locations with point-level inspection of the image, segmentation and metric values.

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Evaluation Datasets and Setup

City	Source	Data	Area	Stages
Jeddah	Private dashcam	1,238 4K videos (20.6 h, 239 GB)	239.4 km^2	1–6
Frankfurt	Cityscapes val	267 annotated images	16.07 km^2	3–6
Paris	MSLS	8,480 georeferenced images	56.94 km^2	3–6

Jeddah

Residential, coastal, mixed-use and historic morphologies; motion blur, variable illumination, 43.1 km of roads.

Hardware

Ubuntu 22.04, Intel i9-10940X, 62 GB RAM, NVIDIA RTX A5000 (24 GB), CUDA 12.2.

Licence Plate Detection: Domain Adaptation is Necessary

YOLOv12n	mAP50	mAP50-95	Precision	Recall
Pretrained	0.002	0.001	0.004	0.037
Fine-tuned (Saudi)	**0.850**	**0.568**	**0.832**	**0.799**
Improvement	+0.848	+0.567	+0.828	+0.762

mAP50 0.002 → 0.850

The pretrained detector fails almost completely on Saudi plates.

Why

Arabic script, different colour schemes and surrounding context. Privacy compliance outside the West requires regional adaptation.

Cross-City Comparison

Mean	Jeddah	Paris	Frankfurt
SVF	**0.480**	0.348	0.033
UVS	**0.204**	0.185	0.125

SVF drives the ranking

Jeddah's wide, open corridors give the highest SVF; with the largest weight, SVF lifts its UVS. Paris: highest GVI (0.223), TRF 0.188.

Dataset effect

Frankfurt: TRF 0.386, HUM 0.022, near-zero SVF - urban canyons and dense pedestrian scenes of the Cityscapes split, not of the city itself.

UVS is sensitive to morphology and data source: report both alongside any cross-city comparison, and do not read it as a universal quality measure.

Conclusion and Future Work

Summary:

1. Raw dashcam video becomes a georeferenced urban-quality map through a fully automated pipeline, without commercial platforms.
2. Region-specific fine-tuning is essential for privacy: Saudi plate mAP50 rises from 0.002 to 0.850.
3. Consistent outputs across Jeddah, Frankfurt and Paris; SVF differences expose the role of urban morphology.

Future work

Fine-tune EoMT on annotated Saudi imagery; extend beyond pixel-ratio proxies (pavement condition, social safety); hexagonal aggregation for neighbourhood-scale analysis.

github.com/mawady/icmv26

Pipeline code, metric definitions and interactive map



Source: Magnific, "Organic Flat Illustration: People Asking Questions."

Questions?

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Appendix - Related Work

Colour correction

White balance (HSV, Lab), Gray World, gamma, histogram equalisation - effect on downstream metrics rarely evaluated.

Semantic segmentation

FCN $\rightarrow$ SegFormer $\rightarrow$ Mask2Former $\rightarrow$ EoMT: a plain ViT matches Mask2Former with lower architectural complexity.

Urban visual indicators

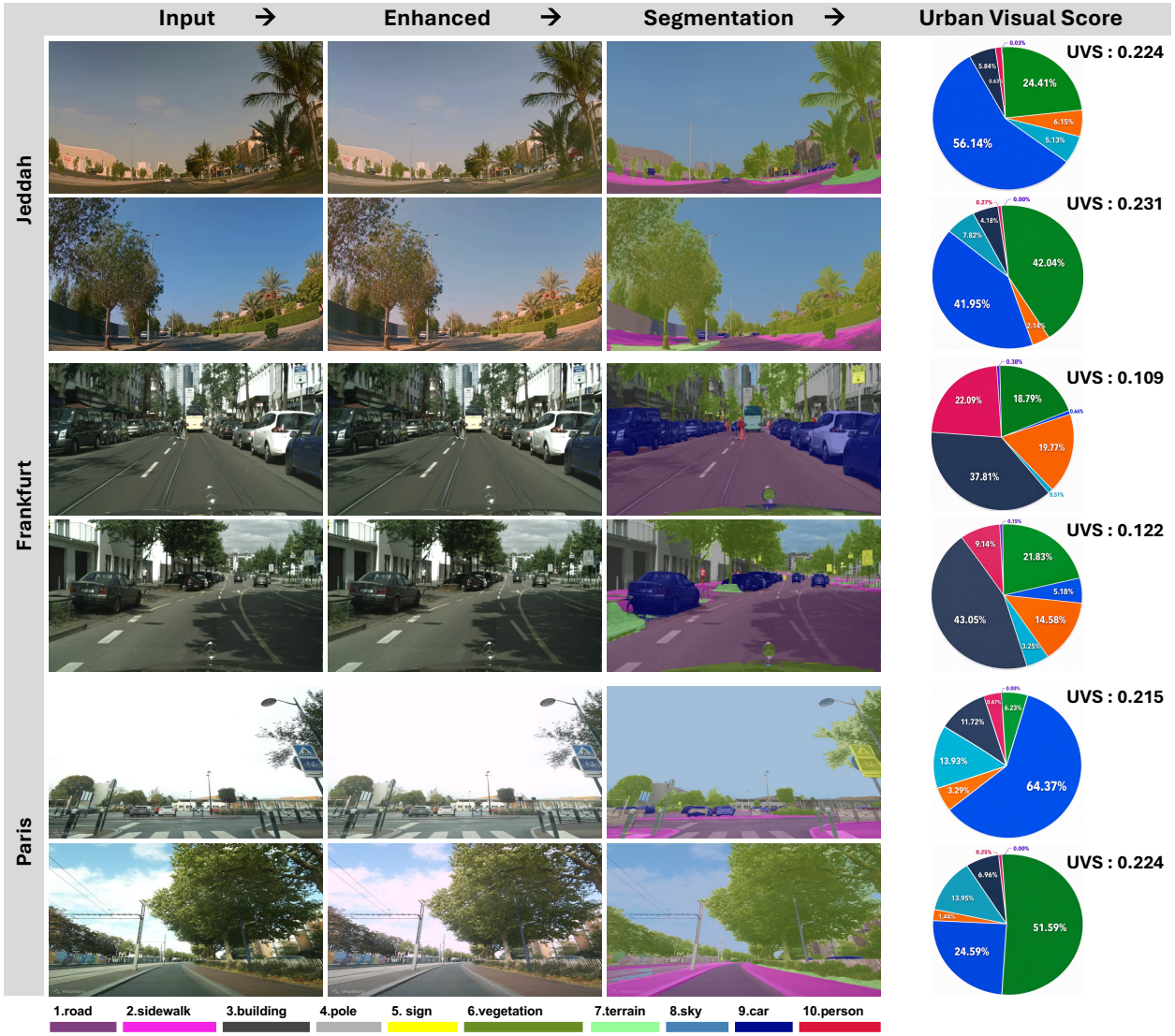
GVI (greenery), SVF (openness), imageability, walkability - usually reported individually.

Urban analysis frameworks

ZenSVI: modular SVI acquisition and analysis. SAGAI: vision-language streetscape assessment. Different inputs and outputs, so not directly comparable.

This work connects raw video, privacy, georeferencing, segmentation and a unified score in one reproducible pipeline.

Appendix - Qualitative Pipeline Outputs



Application Scope

Jeddah
Full pipeline: raw dashcam → mapping.

Frankfurt & Paris
Analysis stages: enhancement → segmentation → metrics.

Appendix - Why These UVS Weights?

Design rationale

GVI and SVF (0.25 each): strongest hypothesised positive contribution to visual quality. WLK (0.20) and IMG (0.15): established but secondary. TRF, TRN, HUM (0.05 each): context-dependent, possibly negative.

What the weights are not

Not empirically optimised or learned from perception data. UVS is a transparent, reproducible heuristic composite, fixed across all cities.

Next step: calibrate against human perception ratings.

Appendix - Limitations

1. Appearance-based proxies

Pixel ratios do not capture functional qualities such as pavement condition or safety.

2. Domain shift in segmentation

EoMT is used without fine-tuning on Jeddah; classes under-represented in Cityscapes may be less accurate.

3. Dataset composition

Cities differ in source and capture method (dashcam vs benchmark vs crowd-sourced), which influences indicator distributions.